SYSTEM ARCHITECTURE



COMPONENTS

COMPONENT	DETAILS
Controller	ESP32 (Wi-Fi) microcontroller in an enclosure near the distribution board. Reports power source, relay states and signal strength to the hub once per second. Relay outputs for the borehole pump and selected heavy loads.
Sensing	Mains voltage and source detection through isolated sensor modules. Ultrasonic level sensor on the overhead tank. (Being added in the next build stage.)
Hub	Runs the automation rules, keeps the event log and serves the dashboard on the local network. Written in Python with no external dependencies; runs on a small always-on device, or a laptop/phone during a pilot.
Dashboard	Web app, nothing to install. Live status, controls, forecasts, event log, Ask A.D.A. Layouts for phone, tablet and wall-mounted screens; per-person access (off, view, edit).
Connectivity	Building Wi-Fi or a dedicated hotspot. The controller always initiates contact with the hub, so no inbound ports are needed on the controller side.

AUTOMATION RULES (DEFAULT SET)

RULE	BEHAVIOUR
Shed on inverter	When the source is inverter, heavy loads (pump, kettle, AC, iron, TV) are switched off and the reason is logged.
Spike protection	Above 260 V, sensitive loads are cut immediately. Safe band shown on the dashboard: 200–250 V.
Pump control	Pump starts when the tank is at or below 20% and NEPA or generator is available; stops at 95%.
Night load control	Selected loads (e.g. freezer) can be scheduled off overnight on inverter and restored when mains returns.
Rules have the last word	A manual command that breaks a rule is reversed, and the dashboard shows which rule and why.

SAFETY

1. **Fail-safe relays:** if the controller cannot reach the hub for 10 seconds, every relay goes to its safe state.
2. **Staleness:** if the dashboard's last reading is more than 5 seconds old, it says so, shows the age, and refuses switch commands. Nothing is queued.
3. **Device offline:** if a controller stops reporting for 5 seconds, the hub marks it offline and logs it.
4. **Electrical work:** all mains connections are made by a qualified electrician.

DATA & ACCESS

1. **Local first:** the event log and readings are kept on the hub, on site.
2. **Device key:** each controller authenticates to the hub with a key generated at setup.
3. **Access control:** each person gets off, view or edit rights per dashboard.
4. **Online assistant:** only when enabled, a short summary of the building's state is sent to answer a question.